Lecture 22 Scribe

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PDF

"Generate a lecture scribe suitable for exam preparation based ONLY on the provided context (Lecture 22 slides). The target audience is a student preparing for a closed-notes exam.

Instructions:

Reason Step-by-Step: While writing the scribe, explicitly explain the reasoning behind definitions, assumptions, proofs, and example solutions exactly as presented in the provided context. Include this reasoning directly in the scribe text to help the student understand 'why' and 'how'.

Faithfulness: Include definitions, notation, theorem statements, and worked examples exactly as they appear in the slides. Do not hallucinate or add information from outside the context.

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Structure: Organize the content with clear headings and subheadings corresponding to the lecture flow.

Goal: The output should allow a student to recall definitions, reconstruct proofs, and follow example solutions step-by-step."

Below is a **lecture scribe for Lecture 22 (Tail Bounds & Jensen's Inequality)** prepared strictly from the provided slides. It is structured for **exam preparation**, with **step-by-step reasoning**, exact formulas, and clear logical flow.

📘 Lecture 22 Scribe: Tail Bounds & Jensen's Inequality

(From provided slides)

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1. Markov Inequality (Foundation)

Definition

For a non-negative random variable X:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Reasoning (Why this works)

- Markov uses only **expectation (mean)**.
- If the mean is small, large values (≥ a) must be rare.
- So it gives a **general but loose bound**.

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2. Chebyshev's Inequality

Statement

$$P(|X - \mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Key Idea (from slide)

- Uses **mean + variance**
- Controls **deviation from mean**

Step-by-Step Derivation (as shown in slides)

1. Start with:

$$P(|X - \mu| \geq a)$$

2. Rewrite:

$$= P((X - \mu)^2 \geq a^2)$$

👉 Reason: Squaring converts absolute deviation into a non-negative quantity.

3. Apply **Markov inequality on** $(X-\mu)^2$:

$$P((X-\mu)^2 \geq a^2) \leq \frac{E[(X-\mu)^2]}{a^2}$$

4. Recognize:

$$E[(X-\mu)^2] = \mathrm{Var}(X)$$

5. Final result:

$$P(|X-\mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

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3. Interpretation of Chebyshev

- Gives bound on "**tail probability" (far from mean)**
- Works for **any distribution**
- Stronger than Markov (because variance adds more info)

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4. Real-Life Example (Exam Scores)

Given (from slide)

- Mean $\mu = 70$
- Std deviation $\sigma = 10$

Question

Fraction of students scoring:

$$< 40 \text{ or } > 100$$

👉 Convert to deviation:

$$|X-70| \geq 30$$

Apply Chebyshev

$$P(|X-70| \geq 30) \leq \frac{10^2}{30^2} = \frac{100}{900} = \frac{1}{9}$$

Final Insight

- At most **1/9 students** are far from mean

---

5. Solved Example 1 (Grades)

Given

- Mean = 70
- Std deviation = 5 → Variance = 25
- Want: $P(X \geq 90)$

Steps (as in slides)

1. Identify:

$$\mu = 70,\ \sigma^2 = 25$$

2. Convert:

$$X \geq 90 \Rightarrow |X - 70| \geq 20$$

👉 Reason: Chebyshev works on deviation from mean

3. Apply inequality:

$$P(X \geq 90) \leq P(|X - 70| \geq 20)$$

4. Use Chebyshev:

$$\leq \frac{25}{20^2} = \frac{25}{400}$$

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6. Solved Example 2 (Network Latency)

Given

- $E[X] = 50$, $\mathrm{Var}(X) = 16$
- Range: [30, 70]

Convert condition

$$X \notin [30, 70] \Rightarrow |X - 50| \geq 20$$

Apply Chebyshev

$$P(|X-50| \geq 20) \leq \frac{16}{400} = \frac{1}{25}$$

Final Result

- At most **4% packets** fall outside range

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7. Chernoff Bound (Key Idea)

Progression (slide)

- Markov → uses X
- Chebyshev → uses $(X-\mu)^2$
- Chernoff → uses e^{tX}

Insight

- Exponential grows **very fast**
- Tail events get **amplified**
- ⇒ gives **tighter bounds**

---

8. Core Trick of Chernoff

Step 1: Transformation

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

👉 Reason:

- Exponential is **monotonic**
- Inequality preserved

Step 2: Apply Markov

$$P(e^{tX} \geq e^{ta}) \leq \frac{E[e^{tX}]}{e^{ta}}$$

👉 Reason:

- $e^{tX} \geq 0$, so Markov applies

Step 3: Final Chernoff Bound

$$P(X \geq a) \leq \min_{t>0} \frac{E[e^{tX}]}{e^{ta}}$$

Key Insight

- Choose best t → **tightest bound**
- Much stronger than Chebyshev

---

9. Lower Tail Chernoff

Goal

$$P(X \leq a)$$

Trick

- Use $t < 0$

$$P(X \leq a) = P(e^{tX} \geq e^{ta})$$

Final Form

$$P(X \leq a) \leq \min_{t<0} \frac{E[e^{tX}]}{e^{ta}}$$

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10. Poisson Tail Bound

Step 1: Compute MGF

$$E[e^{tX}] = e^{\lambda(e^t - 1)}$$

Step 2: Bound

$$P(X \geq a) \leq \min_{t>0} e^{\lambda(e^t - 1) - ta}$$

Optimization

- Minimized at:

$$t = \ln(a/\lambda)$$

Final Result

$$P(X \geq a) \leq e^{a-\lambda} \left(\frac{\lambda}{a}\right)^a$$

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11. Jensen's Inequality

Background

We know:

$$E[X^2] \geq (E[X])^2$$

👉 Question: Does this extend to other powers?

General Statement

If X is positive:

$$E[X^a] \geq (E[X])^a$$

👉 This comes from **convexity**

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12. Convex Function (Definition)

A function $g(x)$ is convex if:

$$g(\alpha x_1 + (1-\alpha)x_2) \leq \alpha g(x_1) + (1-\alpha)g(x_2)$$

👉 Meaning:

- Function lies **below straight line**

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13. From Convexity to Expectation

Generalization

$$g\left(\sum \alpha_i x_i\right) \leq \sum \alpha_i g(x_i)$$

Interpretation

- α_i behave like probabilities
- Sum = 1 → weighted average

---

14. Jensen's Inequality (Final Form)

$$g(E[X]) \leq E[g(X)]$$

Intuition

- Apply function **after averaging → smaller**
- Apply function **before averaging → larger**

👉 Convex functions **push expectation upward**

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15. Simple Example

Given

$$X = \begin{cases} 1 & (0.5) \\ 3 & (0.5) \end{cases}$$

Step 1: Expectation

$$E[X] = 2$$

Step 2: Apply function after

$$g(E[X]) = 2^2 = 4$$

Step 3: Apply function before

$$E[g(X)] = 0.5(1^2) + 0.5(3^2) = 5$$

Conclusion

$$g(E[X]) \leq E[g(X])$$

✔️ Jensen verified

🔑 Final Exam Takeaways

Must Remember:

- Markov → uses mean
- Chebyshev → uses variance
- Chernoff → uses exponential (tightest)

Key Forms:

- Chebyshev:

$$P(|X - \mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

- Chernoff:

$$P(X \geq a) \leq \min_{t>0} \frac{E[e^{tX}]}{e^{ta}}$$

- Jensen:

$$g(E[X]) \leq E[g(X)]$$

If you want, I can also convert this into:

✅ **1-page revision sheet**

✅ **Important MCQs for exam**

✅ **Gujarati explanation (easy memory version)**

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Here is a **faithful, step-by-step lecture scribe for Lecture 22** based *only on the provided slides*. It is structured for **closed-notes exam preparation**, with **explicit reasoning**, **exact formulas**, and **clear logical flow**.

📘 Lecture 22 Scribe: Tail Bounds & Jensen's Inequality

(Source: Lecture 22 slides)

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1. Starting Point: Markov Inequality

Statement

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Reasoning (from slide intuition)

- We only know the **mean** $E[X]$.
- If many values of X were ≥ a, then the mean would be large.
- Since the mean is fixed, the probability of large values must be limited.

👉 This gives a **basic tail bound**, but it is **loose** because it uses only the mean.

---

2. Chebyshev's Inequality

Statement

$$P(|X - \mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Key Idea (slide)

- Uses **mean + variance**
- Controls **deviation from mean**

Proof (Step-by-Step exactly as in slides)

Step 1: Start with deviation

$$P(|X - \mu| \geq a)$$

Step 2: Rewrite using square

$$= P((X - \mu)^2 \geq a^2)$$

👉 Reason:

- Squaring removes absolute value
- Makes expression **non-negative**, so Markov can be applied

---

Step 3: Apply Markov Inequality

$$P((X-\mu)^2 \geq a^2) \leq \frac{E[(X-\mu)^2]}{a^2}$$

👉 Reason:

- Markov applies to any **non-negative random variable**
- Here we use $(X-\mu)^2$

Step 4: Recognize variance

$$E[(X-\mu)^2] = \mathrm{Var}(X)$$

Final Result

$$P(|X-\mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Interpretation (from slides)

- Measures probability of being **far from mean**
- Tail = values far from μ

---

3. Running Example (Coin Flips)

Setup (slide)

- $X \sim \mathrm{Binomial}(n, 1/2)$
- Mean $\mu = n/2$

Event

$$P(X \geq 3n/4)$$

👉 This is a **tail event (far from mean)**

Key Insight (slide)

- Chebyshev gives a **better bound than Markov**

---

4. Real-Life Example: Exam Scores

Given

- Mean = 70
- Std deviation = 10

Question

Fraction of students:

$$\text{below } 40 \text{ or above } 100$$

Step-by-Step Solution

Step 1: Convert to deviation form

$$|X - 70| \geq 30$$

👉 Reason:

- Chebyshev works with distance from mean

Step 2: Apply inequality

$$P(|X - 70| \geq 30) \leq \frac{10^2}{30^2}$$

Step 3: Simplify

$$= \frac{100}{900} = \frac{1}{9}$$

Final Insight (slide)

- At most **1/9 students** are far from mean

---

5. Solved Example 1: Exam Grades

Given

- Mean $\mu = 70$
- Std deviation $\sigma = 5 \Rightarrow \sigma^2 = 25$
- Find upper bound for:

$$P(X \geq 90)$$

Steps (exact slide flow)

Step 1: Identify parameters

$$\mu = 70,\ \sigma^2 = 25$$

Step 2: Convert condition

$$X \geq 90 \Rightarrow |X - 70| \geq 20$$

👉 Reason:

- Chebyshev requires deviation form

Step 3: Apply Chebyshev

$$P(X \geq 90) \leq P(|X - 70| \geq 20)$$

Step 4: Substitute

$$\leq \frac{25}{20^2}$$

Insight (slide)

- Bound is **much tighter than Markov**

---

6. Solved Example 2: Network Latency SLA

Given

- $E[X] = 50$
- $\mathrm{Var}(X) = 16$
- Range: [30, 70]

Step-by-Step Solution

Step 1: Identify parameters

$$\mu = 50,\ \sigma^2 = 16$$

Step 2: Convert failure condition

$$X \notin [30, 70] \Rightarrow |X - 50| \geq 20$$

Step 3: Apply Chebyshev

$$P(|X - 50| \geq 20) \leq \frac{16}{400}$$

Step 4: Simplify

$$= \frac{1}{25}$$

Final Result (slide)

- At most **4% packets fail**

---

7. Chernoff Bound: Key Idea

Progression (slide)

- Markov → uses X
- Chebyshev → uses $(X-\mu)^2$
- Chernoff → uses e^{tX}

Insight

- Exponential grows **very fast**
- Large values become **much larger**
- Tail events get **amplified**

👉 Therefore:

- Leads to **tighter tail bounds**

---

8. Core Trick (Chernoff)

Step 1: Transformation

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

👉 Reason:

- Exponential is **monotonic**
- Inequality preserved

Step 2: Apply Markov

$$P(e^{tX} \geq e^{ta}) \leq \frac{E[e^{tX}]}{e^{ta}}$$

👉 Reason:

- $e^{tX} \geq 0$
- Markov inequality applies

Important Note (slide)

- e^{tX} is always **non-negative**
- Bound now depends on **t**

---

9. Final Chernoff Bound

$$P(X \geq a) \leq \min_{t>0} \left\{ \frac{E[e^{tX}]}{e^{ta}} \right\}$$

Key Insight (slide)

- Works for **all** $t > 0$
- Choose best t → **tightest bound**
- Stronger than Markov & Chebyshev
- Gives **exponential decay**

---

10. Lower Tail Chernoff

Goal

$$P(X \leq a)$$

Trick (slide)

- Use $t < 0$

$$P(X \leq a) = P(e^{tX} \geq e^{ta})$$

Final Bound

$$P(X \leq a) \leq \min_{t<0} \left\{ \frac{E[e^{tX}]}{e^{ta}} \right\}$$

---

11. Poisson Tail Bound

Step 1: Compute expectation

$$E[e^{tX}] = \sum_{i=0}^{\infty} e^{ti} \cdot \frac{e^{-\lambda}\lambda^i}{i!}$$

Step 2: Simplify (as shown in slides)

$$= e^{\lambda(e^t - 1)}$$

Step 3: Apply Chernoff

$$P(X \geq a) \leq \min_{t>0} e^{\lambda(e^t-1)-ta}$$

Step 4: Optimize

- Minimum at:

$$t = \ln(a/\lambda)$$

Final Result (slide)

$$P(X \geq a) \leq e^{a-\lambda} \left(\frac{\lambda}{a}\right)^a$$

---

12. Jensen's Inequality Motivation

Known fact (slide)

$$E[X^2] \geq (E[X])^2$$

Question (slide)

Does it hold for:

$$E[X^3],\ E[X^4],\ E[X^{4.5}]?$$

General Statement

For positive X:

$$E[X^a] \geq (E[X])^a$$

👉 This follows from **Jensen's inequality**

---

13. Convex Function

Definition

A function $g(x)$ is convex if:

$$g(\alpha x_1 + (1-\alpha)x_2) \leq \alpha g(x_1) + (1-\alpha)g(x_2)$$

Understanding (slide)

- Pick two points
- Straight line lies **above curve**

---

14. From Convexity to Expectation

General form

$$g\left(\sum \alpha_i x_i\right) \leq \sum \alpha_i g(x_i)$$

Interpretation

- α_i behave like probabilities
- Sum = 1 → weighted average

Probability interpretation (slide)

- $X = x_i$ with probability $p(x_i)$
- Expectation = weighted average

---

15. Jensen's Inequality (Final Theorem)

$$g(E[X]) \leq E[g(X)]$$

Interpretation (slide)

- Average first → smaller value
- Apply function first → larger value
- Convex functions push expectation **upwards**

---

16. Simple Example

Given

$$X = \begin{cases} 1 & (0.5) \\ 3 & (0.5) \end{cases}$$

Step 1: Compute expectation

$$E[X] = 0.5(1) + 0.5(3) = 2$$

Step 2: Apply function after averaging

$$g(E[X]) = (2)^2 = 4$$

Step 3: Apply function first

$$E[g(X)] = 0.5(1^2) + 0.5(3^2) = 5$$

Conclusion

$$g(E[X]) \leq E[g(X])$$

✔️ Jensen verified

🔑 Final Exam Summary

Inequalities

- Markov:

$$P(X \geq a) \leq \frac{E[X]}{a}$$

- Chebyshev:

$$P(|X - \mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

- Chernoff:

$$P(X \geq a) \leq \min_{t>0} \frac{E[e^{tX}]}{e^{ta}}$$

- Jensen:

$$g(E[X]) \leq E[g(X)]$$

🎯 How to Use in Exam

- Convert problem → **deviation form** for Chebyshev
- Use **exponential trick** for Chernoff
- Recognize **convex function** → apply Jensen

If you want next:
✅ 1-page revision cheat sheet
✅ Important exam MCQs
✅ Gujarati memory tricks

I can prepare that too 👍

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📘 Lecture 22 Scribe: Tail Bounds & Jensen's Inequality

(Source: Lecture 22 slides)

---

1. From Markov to Chebyshev: Motivation

Markov Inequality (Starting Point)

$$P(X \geq a) \leq \frac{E[X]}{a}$$

Reasoning (as implied in slides)

- Uses only **mean** $E[X]$.
- If probability of large values (≥ a) were high, the mean would increase.
- Therefore, the probability must be bounded.

👉 Limitation:

- Does **not consider spread (variance)**
- Hence, gives a **loose bound**

---

2. Chebyshev's Inequality

Statement

$$P(|X-\mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Key Idea (slide)

- Uses **mean + variance**
- Controls **deviation from mean**

Proof (Exact Slide Logic)

Step 1: Start with deviation

$$P(|X-\mu| \geq a)$$

Step 2: Convert absolute value

$$= P((X-\mu)^2 \geq a^2)$$

Why?

- Squaring removes absolute value
- Makes expression **non-negative**
- Needed to apply Markov inequality

Step 3: Apply Markov

$$P((X-\mu)^2 \geq a^2) \leq \frac{E[(X-\mu)^2]}{a^2}$$

Why?

- Markov works for **non-negative random variables**
- Here we use $(X-\mu)^2$

Step 4: Recognize variance

$$E[(X-\mu)^2] = \mathrm{Var}(X)$$

Final Result

$$P(|X-\mu| \geq a) \leq \frac{\mathrm{Var}(X)}{a^2}$$

Interpretation

- Bounds probability of being **far from mean**
- Tail = values far from μ

---

3. Running Example (Coin Flips)

Setup (from slide)

- $X \sim \mathrm{Binomial}(n, 1/2)$
- Mean $\mu = n/2$

Event

$$P(X \geq 3n/4)$$

Insight

- This is a **tail event (far from mean)**

Key Insight (slide)

- Chebyshev gives a **better bound than Markov**

---

4. Real-Life Example: Exam Scores

Given

- Mean = 70
- Std deviation = 10

Question

Fraction of students:

- below 40 OR above 100

Step-by-Step Solution

Step 1: Convert to deviation

$$|X - 70| \geq 30$$

Why?

- Chebyshev works on distance from mean

Step 2: Apply Chebyshev

$$P(|X - 70| \geq 30) \leq \frac{10^2}{30^2}$$

Step 3: Simplify

$$= \frac{100}{900} = \frac{1}{9}$$

Final Insight

- At most **1/9 students** lie far from mean

---

5. Solved Example 1: Exam Grades

Given

- Mean $\mu = 70$
- Std deviation $\sigma = 5 \Rightarrow \sigma^2 = 25$
- Goal: Bound $P(X \geq 90)$

Steps (exact slide flow)

Step 1: Identify parameters

$$\mu = 70, \quad \sigma^2 = 25$$

Step 2: Convert condition

$$X \geq 90 \Rightarrow |X - 70| \geq 20$$

Why?

- Chebyshev requires deviation from mean

Step 3: Apply inequality

$$P(X \geq 90) \leq P(|X - 70| \geq 20)$$

Step 4: Substitute values

$$\leq \frac{25}{20^2}$$

Insight (slide)

- This bound is **much tighter than Markov**

---

6. Solved Example 2: Network Latency SLA

Given

- $E[X] = 50$
- $\mathrm{Var}(X) = 16$
- Range: [30, 70]

Steps

Step 1: Identify parameters

$$\mu = 50, \quad \sigma^2 = 16$$

Step 2: Convert condition

$$X \notin [30, 70] \Rightarrow |X - 50| \geq 20$$

Step 3: Apply Chebyshev

$$P(|X - 50| \geq 20) \leq \frac{16}{400}$$

Step 4: Simplify

$$= \frac{1}{25}$$

Final Result

- At most **4% packets** fall outside range

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7. Chernoff Bound: Motivation

Progression (slide)

- Markov → uses X
- Chebyshev → uses $(X - \mu)^2$
- Chernoff → uses e^{tX}

Key Insight

- Exponential grows **very fast**
- Large values become **much larger**
- Tail events get **amplified**

👉 Hence: **tighter bounds**

---

8. Chernoff Bound: Core Trick

Step 1: Transform probability

$$P(X \geq a) = P(e^{tX} \geq e^{ta})$$

Why?

- Exponential is **monotonic**
- Inequality preserved

Step 2: Apply Markov

$$P(e^{tX} \geq e^{ta}) \leq \frac{E[e^{tX}]}{e^{ta}}$$

Why?

- $e^{tX} \geq 0$, so Markov applies

Important Note (slide)

- e^{tX} is always **non-negative**
- Bound now depends on parameter t

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9. Chernoff Bound (Final Form)

$$P(X \geq a) \leq \min_{t>0} \left\{ \frac{E[e^{tX}]}{e^{ta}} \right\}$$

Key Insights (slide)

- Holds for all $t > 0$
- Choose best t → tightest bound

- Stronger than Markov & Chebyshev
- Gives **exponential decay**

---

10. Lower Tail Chernoff

Goal

$$P(X \leq a)$$

Trick (slide)

Use $t < 0$:

$$P(X \leq a) = P(e^{tX} \geq e^{ta})$$

Final Bound

$$P(X \leq a) \leq \min_{t<0} \left\{ \frac{E[e^{tX}]}{e^{ta}} \right\}$$

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11. Poisson Tail Bound

Step 1: Compute expectation

$$E[e^{tX}] = \sum_{i=0}^{\infty} e^{ti} \cdot \frac{e^{-\lambda}\lambda^i}{i!}$$

Step 2: Simplify (slide derivation)

$$= e^{\lambda(e^t - 1)}$$

Step 3: Apply Chernoff

$$P(X \geq a) \leq \min_{t>0} e^{\lambda(e^t-1)-ta}$$

Step 4: Optimize

Minimum occurs at:

$$t = \ln(a/\lambda)$$

Final Result

$$P(X \geq a) \leq e^{a-\lambda} \left(\frac{\lambda}{a}\right)^a$$

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12. Jensen's Inequality: Motivation

Known fact (slide)

$$E[X^2] \geq (E[X])^2$$

Question (slide)

Does this extend to:

$$E[X^3],\ E[X^4],\ E[X^{4.5}]?$$

General Statement

For positive X:

$$E[X^a] \geq (E[X])^a$$

👉 Comes from convexity

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13. Convex Function

Definition

$$g(\alpha x_1 + (1-\alpha)x_2) \leq \alpha g(x_1) + (1-\alpha)g(x_2)$$

Understanding (slide)

- Pick two points
- Straight line lies above curve

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14. From Convexity to Expectation

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$$g\left(\sum \alpha_i x_i\right) \leq \sum \alpha_i g(x_i)$$

Interpretation

- α_i behave like probabilities
- Sum = 1 → weighted average

Probability view (slide)

- $X = x_i$ with probability $p(x_i)$
- Expectation = weighted average

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15. Jensen's Inequality (Final)

$$g(E[X]) \leq E[g(X)]$$

Interpretation (slide)

- Average first → smaller value
- Apply function first → larger value
- Convex functions push expectation **upwards**

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16. Simple Example

Given

$$X = \begin{cases} 1 & (0.5) \\ 3 & (0.5) \end{cases}$$

Step 1: Expectation

$$E[X] = 2$$

Step 2: Apply function after

$$g(E[X]) = 2^2 = 4$$

Step 3: Apply function first

$$E[g(X)] = 0.5(1^2) + 0.5(3^2) = 5$$

Conclusion

$$g(E[X]) \leq E[g(X)]$$

✔️ Jensen verified

🔑 Final Exam Memory Summary

- **Markov** → uses mean

- **Chebyshev** → uses variance
- **Chernoff** → exponential trick (tightest)
- **Jensen** → convexity + expectation

If you want, I can now make:

✅ **1-page quick revision sheet**
✅ **Important MCQs (exam-focused)**
✅ **Gujarati explanation for fast memory**